ROLL TRIM + ANAMORPHIC CONFORM — PORTAL

Level a drifting solve by eye, conform to 2.39:1, and prove the camera survives the USD round-trip.

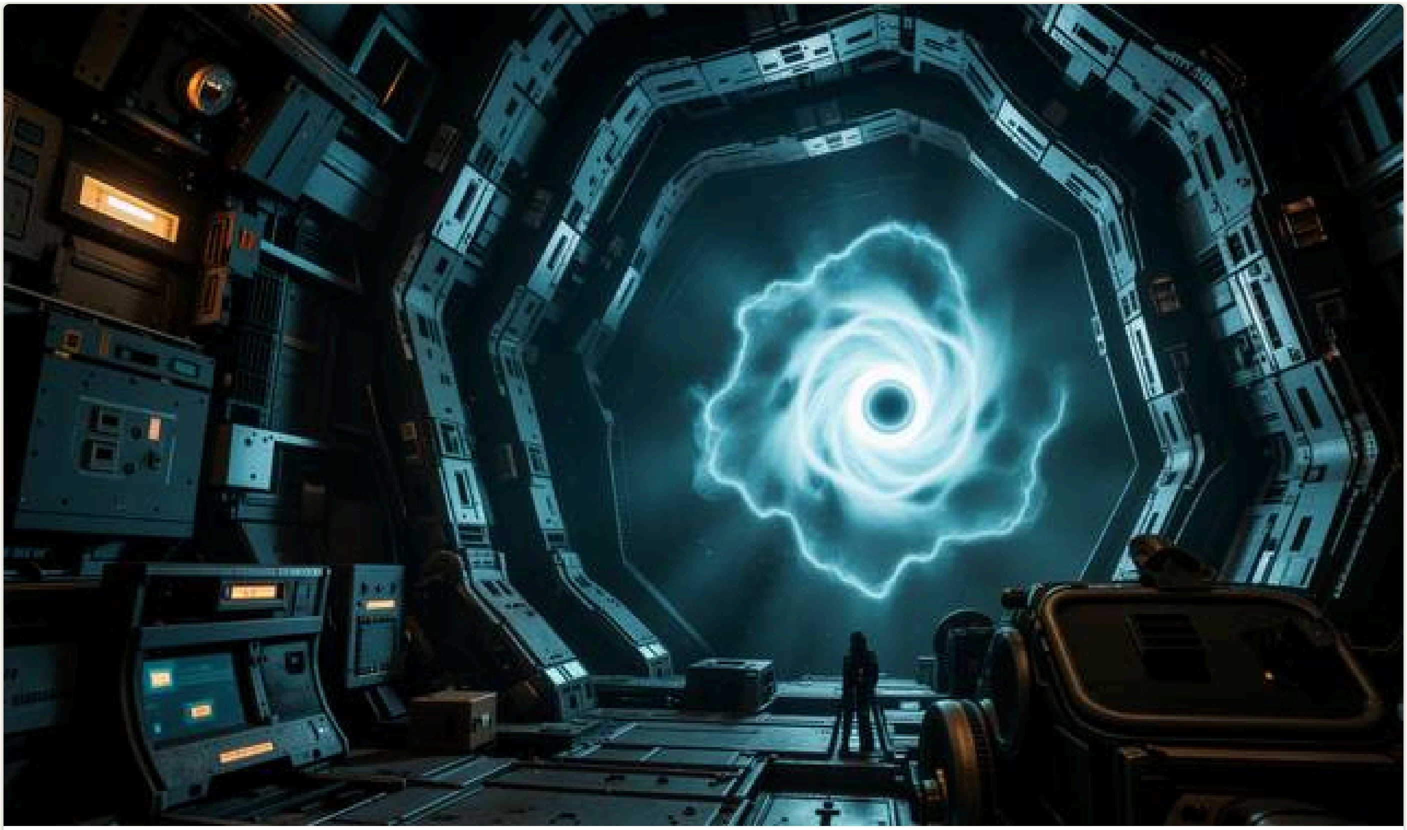
FILE **atlas_rolltrim_shotcam_portal_workflow.json** VERIFIED **2026-07-16**

THE STORY

GeoCalib solved this sci-fi interior's roll at −5.6°, but the architecture's verticals imply about −2.6° — measured against 196 detected edges. Nothing catches this automatically: the classical VP detector found ZERO vanishing points on the greebled octagonal chamber, so there's no signal to auto-correct from. This is exactly what AtlasRollTrim is for: it rotates the recovered camera about its own VIEW AXIS — position and view direction invariant, framing preserved — and you dial the few degrees by eye until the floor grid sits level with the interior lines.

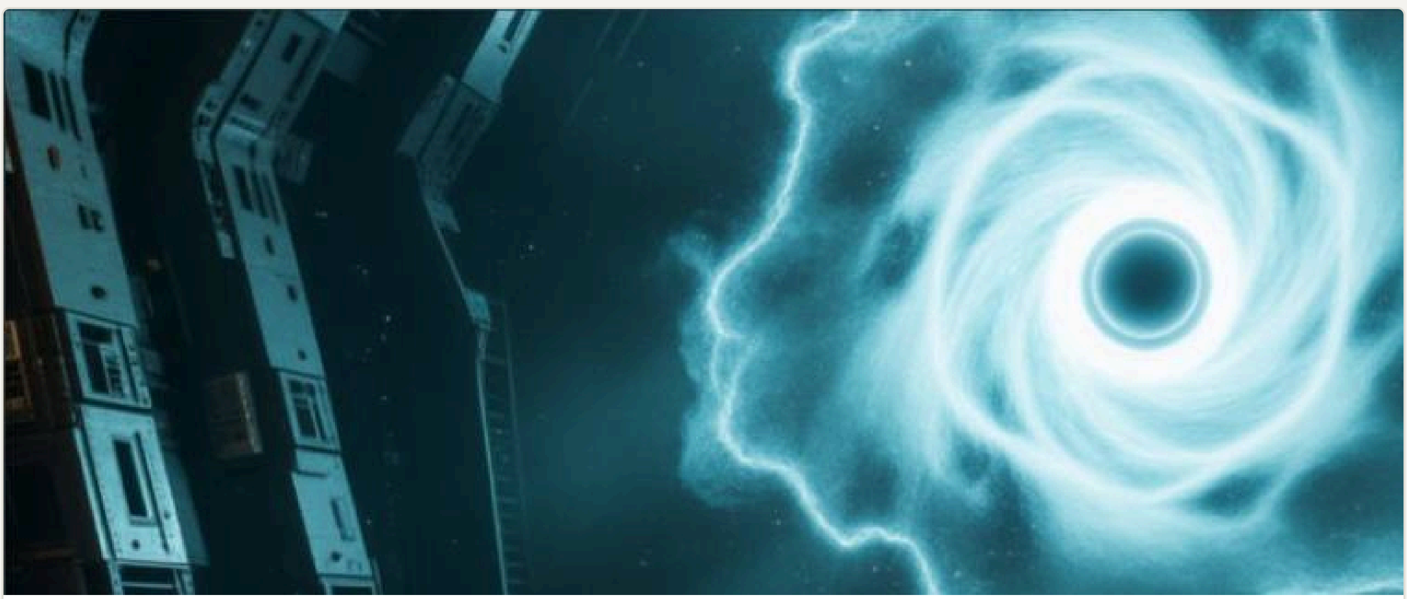
Downstream, a MoGe relief (the interior depth specialist) builds the geometry, a 2.39:1 anamorphic ShotCam conforms the viewport render to a delivery format, and the round-trip proof runs: ExportUSD writes the camera, AtlasUSDCameraLoader reads it back, and AtlasDecomposeCamera confirms the reloaded focal matches the trimmed solve's. The camera survives the DCC handoff intact.

IN THE VIEWPORT — PROJECTED VS. THE GREY MESH



SOURCE PLATE

atlas_00024_portal.png · AI interior · 7680×4512 — the single still photograph everything below is reconstructed from.



PROJECT ON

The photo reassembled on the recovered camera — texels assigned by ray, perfect from this view.

PROJECT OFF

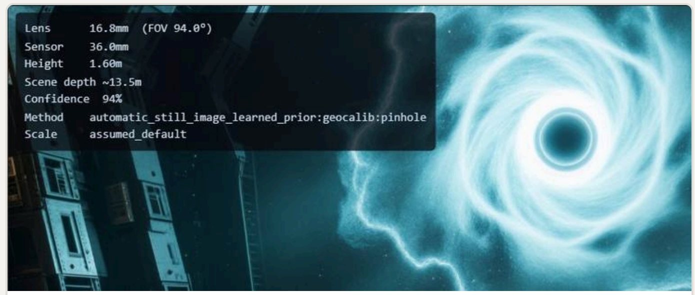
The same geometry as a grey mesh, recovered view — what the depth-derived surfaces actually are.

GREY · ORBIT LEFT

Orbited off-axis: the reconstructed cone. Coverage ends where the camera never saw.

GREY · ORBIT RIGHT

The opposite angle — silhouette tears read as deliberate holes, not errors.



INFO HUD

ℹ overlay: solved lens, sensor, camera height, confidence — the solve's numbers in the corner.

NODES ON STAGE

`AtlasLearnedSolveFromImage` `AtlasRollTrim` `AtlasDepthMap` `AtlasDeriveReliefMesh` `AtlasDefineShotCam` `AtlasExportUSD` `AtlasUSDCameraLoader` `AtlasDecomposeCamera`

RETUNE ON YOUR PLATE

- **roll_deg is entirely per-plate, tuned by eye.** Positive turns the scene counter-clockwise on screen. −3° was this plate's correction; yours will differ.
- **Position in the chain is load-bearing.** RollTrim sits BETWEEN the solve and the derive nodes — geometry back-projects through the corrected view matrix. The node's report warns if geometry already rode in.
- **ShotCam format.** 2.39:1 anamorphic here; set your own sensor/focal/resolution. The letterboxed viewport is the conform working — not a bug.

RUN-VERIFIED

RUN ✓ Viewport conformed to 2048×856 — exactly 2.39:1; the round-tripped USD camera decomposes to the same lens as the trimmed solve.

REQUIREMENTS

`[moge] for the MoGe depth · [usd] for the round-trip`